Example 8.26 If $P(\mathrm{A}) = 0.37$, $P(\mathrm{B}) = 0.42$, $P(A \cap B) = 0.09$ then find $P(A \cup B)$.

Solution $P(\mathrm{A}) = 0.37$, $P(\mathrm{B}) = 0.42$, $P(A \cap B) = 0.09$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B) = 0.37 + 0.42 - 0.09 = 0.7$$

Example 8.27 What is the probability of drawing either a king or a queen in a single draw from a well shuffled pack of 52 cards?

Solution

Total number of cards $= 52$

Number of king cards $= 4$

Probability of drawing a king card $= \frac{4}{52}$

Number of queen cards $= 4$

Probability of drawing a queen card $= \frac{4}{52}$

Thinking Corner

$P(A \cup B) + P(A \cap B)$ is _____.

Both the events of drawing a king and a queen are mutually exclusive

$$\Rightarrow P(A \cup B) = P(A) + P(B)$$

$\therefore$ Probability of drawing either a king or a queen $= \frac{4}{52} + \frac{4}{52} = \frac{2}{13}$

Example 8.28 Two dice are rolled together. Find the probability of getting a doublet or sum of faces as 4.

Solution When two dice are rolled together, there will be $6 \times 6 = 36$ outcomes. Let S be the sample space. Then $n(S) = 36$

Let A be the event of getting a doublet and B be the event of getting face sum 4.

Then $A = \{(1,1),(2,2),(3,3),(4,4),(5,5),(6,6)\}$

$B = \{(1,3),(2,2),(3,1)\}$

$\therefore \quad A \cap B = \{(2,2)\}$

Then, $n(A) = 6$, $n(B) = 3$, $n(A \cap B) = 1$.

$$P(A) = \frac{n(A)}{n(S)} = \frac{6}{36}$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{3}{36}$$

$$P(A \cap B) = \frac{n(A \cap B)}{n(S)} = \frac{1}{36}$$

$\therefore P$ (getting a doublet or a total of 4) = $P(A \cup B)$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{6}{36} + \frac{3}{36} - \frac{1}{36} = \frac{8}{36} = \frac{2}{9}$$

Hence, the required probability is $\frac{2}{9}$.

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